

Pressure-Only ModalPINN Cylinder-Wake Reconstruction

R11 and R12 Results, Interpretation, and the R13 Tighter-Trust Experiment

Research progress summary

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Abstract

This report summarizes the two most recent experiments in the pressure-only ModalPINN cylinder-wake project and defines the next experiment now being run. R11 tested whether longer, more heavily optimized training with 25,000 uniformly distributed physics points could refine the successful R10 radial- v_1 solution. It did not: the wake remained alive, but the fundamental far-wake mode moved away from CFD despite a low PINN loss. R12 then isolated the effect of collocation-point placement by comparing two matched L-BFGS-only runs with the same R10 initialization, the same 25,000 physics points, and the same radial prior, changing only uniform versus wake-biased sampling. Wake-biased sampling clearly improved the reconstructed wake and the full u, v, p fields relative to the matched uniform arm. The remaining issue is that the trained far-wake v_1 is still slightly worse than the raw analytical Kármán prior. R13 therefore tests whether reducing the radial trust radius from $\rho = 0.70$ to 0.65 or 0.60 can preserve more of the accurate Kármán structure while retaining the improvements from wake-biased physics enforcement.

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1 Where the project stood before R11

The inverse problem is to reconstruct the periodic two-dimensional wake of a circular cylinder at $Re = 100$ from only 32 wall-mounted pressure taps, with no velocity measurements in the domain. ModalPINN writes the transverse velocity as

$$v(x, y, t) = v_0(x, y) + 2\Re\left[v_1(x, y)e^{i\omega_0 t} + v_2(x, y)e^{i2\omega_0 t} + \dots\right].$$

The fundamental mode v_1 is the dominant side-to-side vortex-shedding component.

Pressure data plus Navier–Stokes alone repeatedly admitted a low-loss solution with an almost dead wake. The targeted remedy developed before R11 was a radial Kármán trust region applied only to far-wake v_1 :

$$v_1 = S_{v1} + \rho|S_{v1}|\frac{N}{\sqrt{1 + |N|^2}}$$

with a smooth implementation and a hard trusted core downstream. Here S_{v1} is the pressure-derived analytical Kármán street mode and N is the trainable neural-network correction.

The successful R10 smoke run used

$$\rho = 0.70, \quad x/D \geq 3, \quad |y/D| \leq 2,$$

and produced, in the far wake,

$$E_{v1} = 0.2935, \quad \text{corr} = 0.9697, \quad \frac{\|v_1\|}{\|v_1^{CFD}\|} = 0.8275.$$

This established that the radial constraint could stop the pressure-only wake from collapsing.

R10 answered the anti-collapse question. R11 and R12 were designed to answer a harder question: can Navier–Stokes training actually refine the rescued wake rather than simply preserve an analytical scaffold?

2 R11: longer uniform-physics refinement

2.1 R11 design

R11 warm-started from the successful R10 radial checkpoint and kept the same radial configuration,

$$\rho = 0.70, \quad x_{start} = 3D.$$

The training budget was increased to

$$N_{int} = 25,000, \quad N_{mes} = 5,000,$$

with four collocation grids, up to 6,000 L-BFGS iterations, up to 7,500 function evaluations, and an Adam stage within a total target of about 1.5 hours. The purpose was to test whether additional physics optimization would move the R10 solution closer to CFD and ideally make it better than the raw Kármán prior.

2.2 R11 fundamental-mode result

Table 1: R11 v_1 result compared with the R10 starting point and the raw Kármán prior.

Case	Relative L_2 error	Complex correlation	Amplitude ratio
R10 start, far wake	0.2935	0.9697	0.8275
R11 final, far wake	0.3913	0.9438	0.7378
R10 start, far core	0.2867	0.9721	0.8259
R11 final, far core	0.3865	0.9466	0.7362
Raw Kármán prior, far core	0.2819	0.9775	0.8082

Relative to the R10 far-wake starting point, the R11 v_1 error increased by approximately

$$\boxed{33.3\%},$$

while the amplitude ratio fell by about 10.8%. The wake did not die, but it became less accurate.

R11 verdict: NO_GO_WAKE_DEGRADED. Longer uniform training did not refine the far-wake v_1 ; it moved it away from CFD.

2.3 R11 full-field accuracy

R11 was not a useless solution. Some full-field errors were reasonable, especially close to the cylinder.

Table 2: R11 regional relative L_2 errors for the reconstructed fields.

Region	E_u	E_v	E_p
Near cylinder	0.0845	0.0741	0.0259
Near wake	0.1875	0.5453	0.3253
Far wake	0.2540	0.5802	1.2902
Upstream/off-axis	0.0482	0.1912	0.2830
Whole domain	0.1968	0.4499	0.3120

The final reported loss components were

$$L_{total} = 9.66 \times 10^{-4}, \quad L_{physics} = 9.37 \times 10^{-4}, \quad L_{pressure} = 2.90 \times 10^{-5}.$$

Thus, the PINN could obtain a small physics-plus-pressure objective while the wake mode became less accurate.

R11 strengthened the identifiability lesson: a lower PINN objective is not sufficient evidence that the pressure-only reconstructed wake is moving toward the true CFD solution.

2.4 Why R11 led to R12

R11 changed several things at once: more uniform collocation points, longer L-BFGS, and Adam. The result therefore did not isolate whether the problem was the spatial physics-point distribution, over-optimization, or the later Adam stage. The next experiment was designed to remove those confounds.

3 R12: matched test of collocation-point placement

3.1 R12 experimental design

R12 used two matched arms. Both began from the exact same successful R10 checkpoint and used:

- 32 pressure taps only;
- the same Navier–Stokes equations and boundary conditions;
- the same radial v_1 prior with $\rho = 0.70$ and $x_{start} = 3D$;
- $N_{int} = 25,000$ and $N_{mes} = 5,000$;
- the same seed and the same network;

- L-BFGS only, with up to 6,000 iterations and 7,500 evaluations;
- no Adam.

The only experimental variable was the spatial distribution of the 25,000 physics points.

Table 3: Wake-biased source mixture used in R12.

Sampling component	Fraction	Points
Whole-domain uniform	30%	7,500
Formation wake: $0 \leq x/D < 3, y/D \leq 2$	35%	8,750
Far wake: $3 \leq x/D \leq 8, y/D \leq 2$	25%	6,250
Near-cylinder annulus: $0.5 < r/D \leq 1.2$	10%	2,500

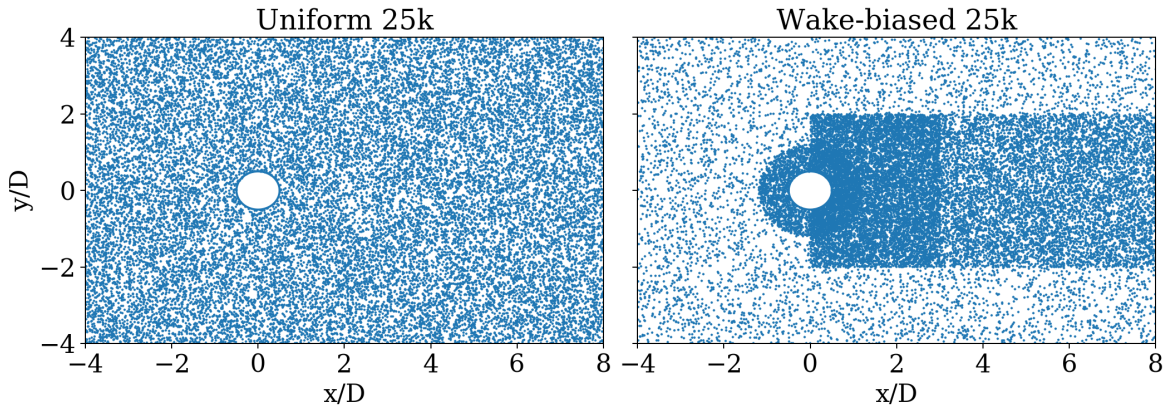


Figure 1: R12 sampling audit: uniform collocation (left) versus the wake-biased mixture (right). The total number of points is identical.

The wake-biased sampling is intentionally *not* importance-corrected. It therefore changes the training objective by putting more Navier–Stokes weight in the physically difficult wake and vortex-formation region.

3.2 R12 v_1 result: wake bias clearly helped

Table 4: Matched R12 fundamental-mode comparison.

Case	E_{v_1}	Correlation	Amplitude ratio
Uniform, far wake	0.3856	0.9447	0.7454
Wake-biased, far wake	0.3119	0.9661	0.8084
Uniform, far core	0.3805	0.9476	0.7438
Wake-biased, far core	0.3035	0.9692	0.8068
Raw Kármán prior, far core	0.2819	0.9775	0.8082

Wake-biased collocation reduced the matched far-wake v_1 error by

$$\boxed{19.1\%}$$

and the far-core v_1 error by

$$\boxed{20.2\%}$$

relative to uniform collocation.

R12 verdict: STRONG_PASS_WAKE_BIAS_HELPED. With the same number of physics points, putting them in more useful locations produced a substantially better wake solution.

3.3 R12 full-field result

The improvement was not limited to the modal diagnostic. The reconstructed full fields also improved strongly in the far wake.

Table 5: R12 regional full-field comparison. Bold values are better.

Region	Sampling	E_u	E_v	E_p
Near cylinder	Uniform	0.1310	0.0954	0.0442
	Wake-biased	0.0881	0.1007	0.0237
Near wake	Uniform	0.1932	0.5291	0.3290
	Wake-biased	0.1787	0.5449	0.3623
Far wake	Uniform	0.2514	0.5790	1.3551
	Wake-biased	0.2079	0.4404	0.8849
Whole domain	Uniform	0.1998	0.4468	0.3246
	Wake-biased	0.1719	0.3854	0.2692

In the far wake, wake-biased sampling reduced

$$E_u \text{ by } 17.3\%, \quad E_v \text{ by } 23.9\%, \quad E_p \text{ by } 34.7\%.$$

Whole-domain errors also improved by approximately 13.9% in u , 13.7% in v , and 17.0% in p .

3.4 Independent fixed-uniform physics audit

The raw training physics losses cannot be directly compared between the two R12 arms because they sample different probability distributions. R12 therefore evaluated both final models on the same independent set of 20,000 uniformly sampled space-time physics points.

Table 6: R12 fixed-uniform Navier–Stokes residual audit. Lower is better.

Audit region	Uniform-trained model	Wake-biased model
Whole domain	3.941×10^{-3}	2.343×10^{-3}
Formation wake	3.084×10^{-3}	2.980×10^{-3}
Far core	1.082×10^{-3}	2.373×10^{-3}
Near cylinder	8.448×10^{-2}	5.586×10^{-3}
Outside wake	5.000×10^{-3}	2.214×10^{-3}

The wake-biased model reduced the *whole-domain* uniform-audit NS residual by approximately

$$\boxed{40.5\%}.$$

This is important because the model did not simply improve the wake by neglecting physics everywhere else. In fact, the near-cylinder audit residual fell by more than an order of magnitude.

The notable exception is the far-core residual, which increased from 1.08×10^{-3} to 2.37×10^{-3} even though the actual far-wake reconstruction became much more accurate. This is another warning against using a local PDE residual alone as a proxy for inverse reconstruction accuracy.

4 What R12 solved, and what remains unsolved

R12 gives a clear positive conclusion:

Collocation-point placement matters strongly for this pressure-only inverse problem.

With the same 25,000-point budget, wake-biased physics enforcement gave a much better full wake than uniform physics enforcement.

However, two limitations remain.

4.1 The best trained v_1 is still not better than the raw Kármán prior

The R12 wake-biased far-core result is

$$E_{v_1} = 0.3035,$$

whereas the raw Kármán prior gives

$$E_{v_1} = 0.2819.$$

The trained mode is therefore still about 7.7% worse in relative error than the analytical scaffold it starts around.

4.2 Continuing from R10 still slightly deforms a very good v_1

The original R10 far-wake result was

$$E_{v_1} = 0.2935,$$

whereas R12 wake-biased training ended at

$$E_{v_1} = 0.3119.$$

Thus, wake-biased sampling is much better than uniform continuation, but the longer optimization still moves v_1 slightly away from its unusually good R10 state.

The amplitude itself is no longer the main problem. R12 gives

$$\frac{\|v_1\|}{\|v_1^{CFD}\|} = 0.8084,$$

which is almost identical to the raw Kármán prior amplitude ratio of 0.8082. The remaining mismatch is primarily in detailed complex shape/phase rather than simple global amplitude.

The next question is no longer “how do we stop the wake dying?” It is now “how much correction freedom should the PINN be allowed around an already accurate far-wake Kármán scaffold?”

5 R13: what we are testing now

5.1 R13 hypothesis

The radial correction has nominal radius

$$\rho|S_{v1}|.$$

At $\rho = 0.70$, the network has substantial freedom to move around the analytical Kármán mode. R13 tests whether that freedom is still too large.

The two new cages are

$$\boxed{\rho = 0.60} \quad \text{and} \quad \boxed{\rho = 0.65},$$

which are compared with the already-completed matched R12 reference

$$\boxed{\rho = 0.70}.$$

Lower ρ means a tighter correction disk around S_{v1} . Nominally, the maximum correction is reduced from 70% of the prior magnitude to 65% or 60%, while the anti-collapse property remains because all three values satisfy $\rho < 1$.

5.2 Fair experimental controls

All three ρ values are compared under the same essential setup:

- same successful R10 latent-weight checkpoint;
- same 32 pressure taps;
- same pressure-derived Kármán prior;
- same hard trusted region downstream of $x/D = 3$;
- same wake-biased 25,000-point collocation mixture from R12;
- same seed;
- same L-BFGS limits: 6,000 iterations and 7,500 evaluations;
- no Adam.

Only $\rho = 0.60$ and 0.65 need new training. The $\rho = 0.70$ result is reused directly from the matched R12 wake-biased arm.

5.3 Pre-training cage-feasibility test

Before spending GPU time, R13 evaluates each cage against the CFD v_1 solely as a diagnostic. It measures:

- the fraction of far-core CFD points inside the allowed complex disk;
- the fraction of true v_1 energy located at reachable points;
- the irreducible relative L_2 error imposed by the cage geometry itself.

This ensures that a tighter value such as $\rho = 0.60$ is not selected merely because it protects the prior while accidentally excluding too much of the real solution.

5.4 Zero-training remap

Changing ρ changes the output mapping even if the underlying R10 neural-network weights are untouched. R13 therefore evaluates the exact same R10 latent weights at

$$\rho = 0.60, \quad 0.65, \quad 0.70$$

before new optimization. This separates the immediate geometric effect of a tighter cage from what L-BFGS subsequently learns.

5.5 R13 decision metrics and guardrails

The primary ranking metric is the far-core v_1 relative error. A tighter cage is only considered acceptable if it also passes all of the following guardrails:

1. far-wake amplitude ratio ≥ 0.65 ;
2. far-wake complex correlation ≥ 0.95 ;
3. full far-wake E_v is not more than 0.03 worse than the $\rho = 0.70$ R12 reference;
4. whole-domain fixed-uniform NS audit is not more than 30% worse than the $\rho = 0.70$ reference.

Among candidates that pass those guardrails, R13 selects the ρ with the lowest far-core v_1 error.

Table 7: R13 decision logic.

Outcome	Interpretation
TIGHTER_RHO_CLEAR_WIN	$\rho = 0.60$ or 0.65 gives at least 0.02 absolute improvement in far-core v_1 error relative to 0.70 while passing all guardrails.
TIGHTER_RHO_SMALL_WIN	A tighter cage is best, but the gain is smaller than 0.02 absolute.
RHO_070_REMAINS_PREFERRED	Neither tighter cage provides a safe improvement; keep the R12 $\rho = 0.70$ formulation.

6 Current research interpretation

The experimental sequence now has a coherent logic:

1. **R10:** radial far-wake v_1 trust solved the catastrophic wake-collapse problem.
2. **R11:** simply optimizing longer with uniform physics points did not refine the wake and could degrade v_1 despite a low objective.
3. **R12:** redistributing the same physics-point budget toward the wake gave a strong matched improvement in v_1 , the full far-wake fields, and the independent whole-domain physics audit.
4. **R13:** now tunes how much freedom the network should have around the analytical far-wake scaffold, testing $\rho = 0.60$ and 0.65 against the established 0.70 reference.

The project has moved from “can pressure-only ModalPINN recover any wake at all?” to a much more focused design problem: find the smallest physically safe Kármán trust radius and the most informative physics-point distribution that jointly preserve identifiability while allowing Navier–Stokes refinement.

The full 50,000-point production run should therefore wait until R13 identifies the preferred ρ .

Data basis for this report

The numerical values in this document were taken directly from the completed experiment outputs:

- R11 outputs: `R11_medium_verdict.json`, `R11_medium_v1.json`, `R11_medium_regions.json`, and the training-loss summary;
- R12 outputs: `R12_verdict.json`, both v_1 and regional evaluation files, and both fixed-uniform physics-audit files;
- the current `R13_tighter_rho_wakebiased_T4.ipynb` definition for the R13 plan and guardrails.

No R13 numerical training result is claimed here because R13 is the experiment currently being run.